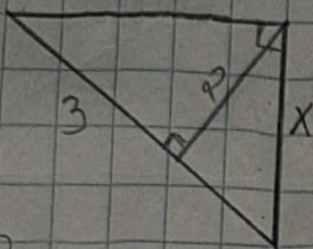


Hoja de Trabajo 3

Ejercicio 1.

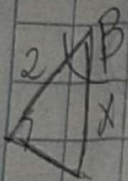


$$C = \sqrt{3^2 + 2^2}$$

$$C = \sqrt{13}$$

$$\frac{\sin 90^\circ}{\sqrt{13}} = \frac{\sin \alpha}{2}$$

$$\alpha = 33.69$$



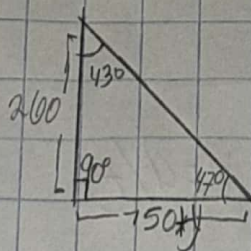
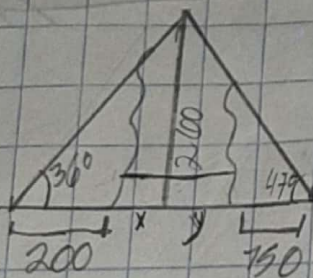
$$\beta = 180 - 90 - 33.69 = 56.31$$

$$90 - 56.31 = 33.69$$

$$180 - 90 - 33.69 = 56.31$$

$$\frac{\sin 56.31}{2} = \frac{\sin 90^\circ}{x} = x = 2.38$$

Ejercicio 2.

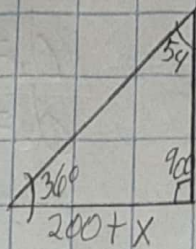


$$\frac{\sin 47^\circ}{260} = \frac{\sin 43^\circ}{150+y}$$

$$y = \frac{260 \sin 43^\circ}{\sin 47^\circ} - 150 = 92.45$$

$$157.86 + 92.45 = 250.31$$

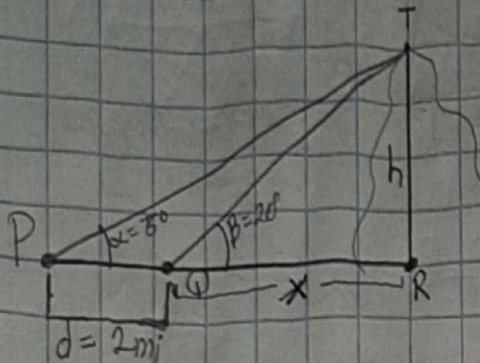
La longitud del túnel es de 250 pies



$$\frac{\sin 36^\circ}{260} = \frac{\sin 54^\circ}{200+x}$$

$$x = \frac{260 \sin 54^\circ}{\sin 36^\circ} - 200 = 157.86$$

Ejercicio 3.



$$\tan 20 = \frac{h}{x} \quad x = \frac{h}{\tan 20}$$

$$x = 2.75h$$

$$\tan 15 = \frac{h}{x+2} = \frac{h}{2.75h+2} \quad h = \tan 15(x+2)$$

$$h = 0.27x + 0.54$$

$$h = 0.27(2.75h) + 0.54$$

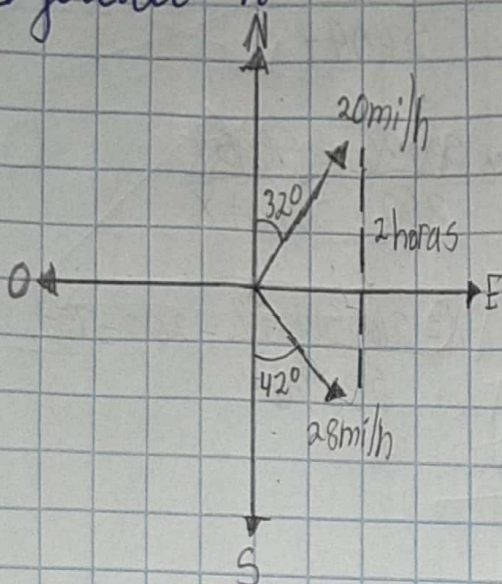
$$h = 0.74h + 0.54$$

$$0.26h = 0.54$$

$$h = 2.08$$

La altura de la montaña
es de 2.08 mi

Ejercicio 4.



$$v = d/t \quad d = vt$$

$$d_1 = 20(2) = 40 \text{ mi}$$

$$d_2 = 28(2) = 56 \text{ mi}$$

$$d = \sqrt{56^2 + 40^2 - 2(56)(40)\cos 70^\circ}$$

$$d = 77.30$$

Después de 2 horas los barcos
están a 77.3 mi